AI FOR GOOD · HACKATHON 7 · EYES ON EARTH

SlootWacht

AI-powered monitoring of Dutch ditches

SDG 15 · Life on Land

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THE PROBLEM

Dutch freshwater is under pressure

1 of 741

assessed water bodies reached good overall ecological status (≈0.13%)

≈85%

had moderate, poor or bad biological quality

Source: CLO, Water Framework Directive assessment 2024. These figures cover assessed water bodies, not every ditch – and SlootWacht cannot measure them directly. They show why monitoring matters.

WHO IT IS FOR

Too many kilometres to inspect by eye

20,507 km

of waterways managed by just one water authority (Hoogheemraadschap Hollands Noorderkwartier, 2024)

OUR USER

A field inspector at a waterschap

Decides which locations need a closer look or maintenance. 21 water authorities, 16,712 employees in the sector (HR Monitor 2025).

Not every kilometre needs the same inspection – but the scale shows why help with sorting photos is useful.

OUR SOLUTION

From ditch photo to inspection report

1. Upload

Inspector uploads a photo of a ditch

2. Detect

Roboflow model draws boxes with confidence scores

3. Prioritise

Overview per class plus an indicative priority

4. Report

Downloadable inspection report

Prototype: the priority is a simple rule, not a validated ecological score, and we have not yet measured time saved.

WHAT IT DETECTS

Three classes, chosen for inspectors

drijvend_afval

Floating litter

Plastic packaging, bottles and other waste in the water. Usually needs removal.

dichte_waterplant

Dense vegetation

Including duckweed. An observation, not a problem: plants are habitat too.

brug_afvoer

Bridges & culverts

Drainage openings where blockages can affect water flow.

Each class marks a place worth a look – the inspector decides what, if anything, needs to happen.

HOW WE BUILT IT

308 labelled images, one pretrained model

Dataset SlootWacht_v1

Scraped with Python from Wikimedia Commons and iNaturalist, then reviewed and re-annotated by hand in Roboflow.



Train 222 (72%) **Valid** 58 (19%) **Test** 28 (9%)

Model: RF-DETR Small

- Pretrained on Objects365
- Images resized to 512 × 512
- 100 training epochs
- 3 detection classes

App: Python + Streamlit, calling the trained model through the Roboflow API.

RESULTS

How well does it detect?

78.1%

Precision

Of all boxes drawn, how many were right

57.0%

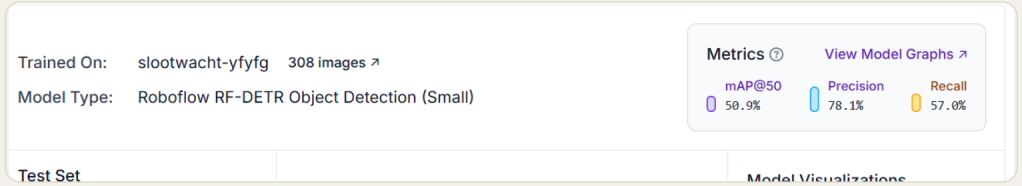
Recall

Of all real objects, how many were found

50.9%

mAP@50

Overall detection quality across classes



Most boxes are right, but it misses about 4 in 10 objects – so no box never means a clean ditch.

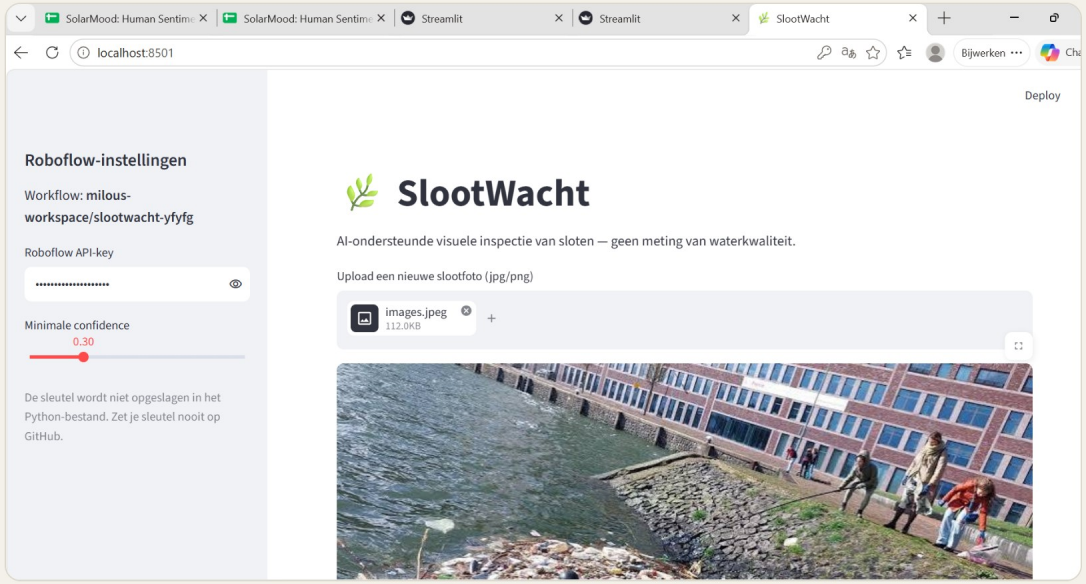
LIVE DEMO

Let’s inspect a ditch

Upload a new photo → see the boxes → read the priority → download the report

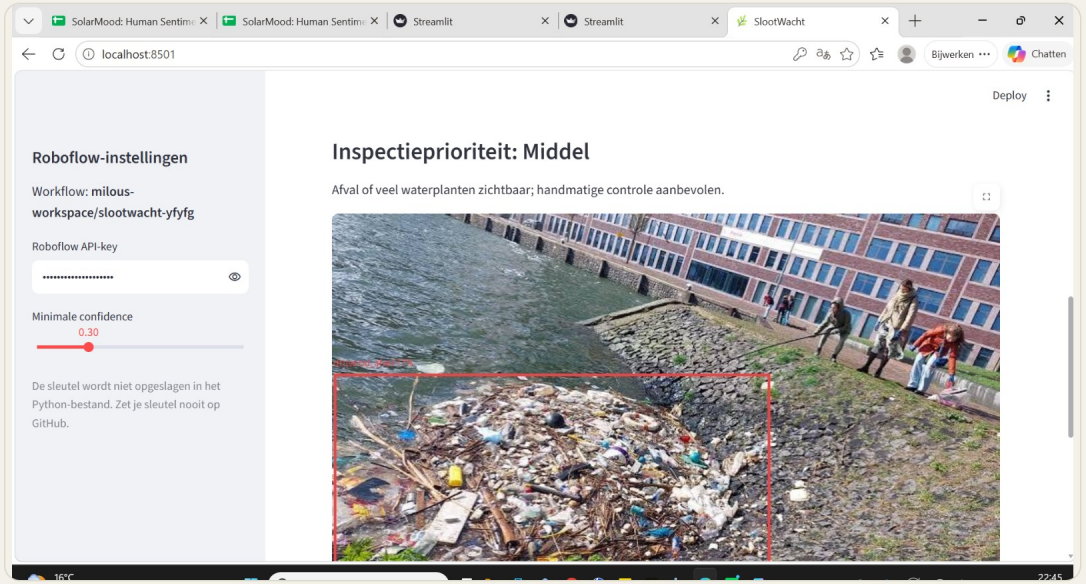
PROOF IT WORKS

A new photo, a correct detection



1. Upload an unseen photo

The app runs locally and calls our trained Roboflow workflow.



2. Litter detected, priority “Middel”

The model boxes the litter as drijvend_afval and advises a manual check.

ETHICAL REFLECTION

What could go wrong

Missed litter

No detection is not proof of a clean ditch. Trusting it blindly lets waste stay.

Removing healthy plants

Flagged vegetation could trigger needless mowing – harming the biodiversity SDG 15 protects.

Biased dataset

308 online photos, some not Dutch, mixed seasons. Real field photos may score worse.

Not a water-quality test

It sees objects, not nitrogen, oxygen, pesticides or toxicity. Plus privacy and image licences.

Our rule: SlootWacht never orders maintenance. A qualified person always decides.

WHAT WE LEARNED

Most of the work was the data

What was hard

- Few photos show litter in a ditch – many showed it on land
- HTTP 429 errors and timeouts while scraping
- Auto-generated boxes needed manual fixing
- Python 3.14 broke Roboflow's inference package

How we solved it

- Other sources, new search terms, smaller batches
- Reviewed every image and box by hand
- Rebuilt the app on Streamlit, Pillow and Requests
- Learned: relevant data beats more data

308 images – just 2.7% above the minimum. Next step: our own photos of Dutch ditches across seasons.

AI FOR GOOD, IN ONE LINE

Technology should support ecological judgement – not replace it.

Questions?

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